

Technology Stack

Recommended implementation stack

Date	30 September 2026	
Team ID	SWDIT-2026-9380	
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models	
Team	Lavinraj (Team Leader), Subash, Rahul, Hariharansudhan	
Maximum Marks	2 Marks	

Layer	Technology	Purpose / justification
Frontend	React + TypeScript + semantic CSS	Reusable form and plan components, strong typing, responsive experience.
Backend	Python FastAPI or TypeScript server functions	Validated server-side requests and protected model credentials.
AI service	Google Gemini API; configurable current Gemini model	Strong instruction following and structured text/JSON generation. Model name stays configurable to avoid lock-in.
Validation	Pydantic or Zod + JSON Schema	Validates user input and generated plan shape before display.
Storage	PostgreSQL (optional for accounts/history)	Reliable relational storage for profiles, plan versions and feedback.
Testing	Pytest/Vitest, Playwright, k6	Unit, interface and performance validation.
Version control	Git + GitHub	Team collaboration, review and automated checks.
Deployment	Managed HTTPS application hosting	Secure deployment, environment variables and scalable request handling.

AI configuration principles

- Keep the Gemini model identifier in configuration rather than hard-coding it throughout the application.
- Request structured JSON and validate it before rendering.
- Use a low-to-moderate temperature for consistent plans.
- Set request timeout, output limits and a user-safe failure state.
- Never place the Gemini API key in browser code or committed files.